Chapter 1

Numerical sequences

Competences: search, model, represent, calculate, communicate

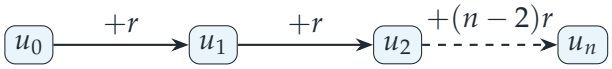
1.1 Reference course

A numerical sequence combines each natural integer n a number u_n. The index n means a rank; it does not must not be confused with the value u_n.

1.1.1 Arithmetic Sequence

Key ideas

A sequence (u_n) is **arithmetic** where there is a real r such as, for all n, $u_{n+1} = u_n + r$. The number r is the common difference / ratio. If the first term is u_0, then $u_n = u_0 + nr$. If the first term is u_1, then $u_n = u_1 + (n-1)r$.



Three numbers x, y, z, written in this order, are three consecutive terms of an arithmetic sequence if $y - x = z - y$, i.e. if $y = \dfrac{x+z}{2}$. This last quantity is the arithmetic mean of x and z.

Method

To model an additive situation: 1) define u_n and specify the unit; 2) identify the first term; 3) translate constant increase or decrease by common difference / ratio r ; 4) Write the term 5) answer with a contextualized sentence.

Worked example

A business sells 480 units the first month and then 25 more units each month. $u_1 = 480$ and $r = 25$, we get $u_n = 480 + 25(n-1)$. Thus $u_{12} = 755$: trade provides 755 units sold in the twelfth month.

1.1.2 Sum of arithmetic terms

For $n+1$ terms of u_0 to u_n,

$$u_0 + u_1 + \cdots + u_n = (n+1)\frac{u_0 + u_n}{2}.$$

The formula is stored in the form of: number of terms multiplied by the average of the first and last term .

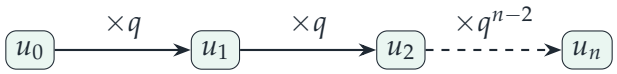
Frequent error

In $u_3 + u_4 + \cdots + u_{12}$, there are $12 - 3 + 1 = 10$ terms, not 12.

1.1.3. Geometric Sequence

Key ideas

A sequence (u_n) is **geometrical** where there is a real q such that $u_{n+1} = qu_n$. If the first term is u_0, then $u_n = u_0 q^n$. If the first term is u_1, then $u_n = u_1 q^{n-1}$.



An increase of t % corresponds to $q = 1 + \dfrac{t}{100}$; a decrease of t % à $q = 1 - \dfrac{t}{100}$. Three positive terms x, y, z are consecutive in a geometric sequence if $y^2 = xz$; $y = \sqrt{xz}$ is their geometric mean.

For $q \neq 1$,

$$u_0 + u_1 + \cdots + u_n = u_0 \frac{1 - q^{n+1}}{1 - q}.$$

1.1.4 Algorithmic: calculate a sum

```
def sum_squares(n):
    s = 0
    for k in rows (1, n + 1):
        s = s + k**2
    return s
```

Accumulator `s` contains the partial sum; the counter `k` travels the entires from 1 to n.

Session 1 — Arithmetic sequence: recognize and model

Numerical sequences

15/09/2026 • B : 2 h – C : 1 h

Objectives of the session

· Recognize an additive evolution.

· Determine the common difference / ratio.

· Define a sequence in a context.

Fluency · 6 to 8 minutes, without calculator

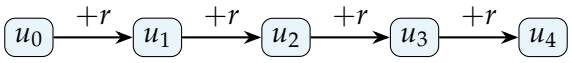
1. Write the multiplier coefficient of a 10 per cent increase. $1 + \frac{10}{100} = 1,10$
2. Calculate u_4 if $u_0 = 14$ and $u_{n+1} = u_n + 3$. $u_4 = 26$
3. How many terms of u_3 to u_{15} ? $15 - 3 + 1 = 13$
4. Give the coefficient of a decrease of 8 per cent. 0,92
5. Give common difference / ratio for $12, 12 + 4, 12 + 2 \times 4, \ldots$. $r = 4$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: recognize an additive evolution and determine the Right.

Research activity

An association has 320 members and plans to win 18 each year. u_n the number of staff after n years. u_0, u_1, u_2. What operation is moving from one term to the next? Write a relationship between u_{n+1} and u_n, then plan u_5.



Detailed correction · Professor edition

$u_0 = 320$, $u_1 = 338$, $u_2 = 356$; add 18. $u_{n+1} = u_n + 18$ and $u_5 = 410$. Have the data that triggers the choice of method.

Trace written

A sequence is arithmetic when $u_{n+1} - u_n = r$ is constant. In a situation, an increase absolute constant leads to an arithmetic sequence.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$u_0 = 320$, $u_1 = 338$, $u_2 = 356$; add 18. $u_{n+1} = u_n + 18$ and $u_5 = 410$.

Training · essential

1. Next steps (u_n) checks $u_0 = 110$ and $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.
2. A quantity is worth 420 and a decrease of 3 per cent per period. v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Detailed correction · Professor edition

$u_n = 110 + 3n$ and $u_{12} = 146$, For the second, $q = 0,97$ and $v_n = 420 \times q^n$; the third item is justified by the consistency of the difference or quotient,

Reference in the manual

Supplements: Calao, courses p. 18·21; exercises p. 24, n° 1·3.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 24, exercise 5

For each of the sequences, specify whether it is arithmetic and, if so, indicate its common difference.

a. For the integer n : $u_n = 2(n-5) + 3n$.

b. For the integer n : $v_n = n^2 + n - 1$.

c. For the integer n : $w_n = -4 + 3n$.

Detailed correction · Professor edition

a. $u_n = 5n - 10$, so $u_{n+1} - u_n = 5$: the sequence is arithmetical of common difference 5.

b. $v_{n+1} - v_n = 2n + 2$, which depends on n : the sequence is not arithmetic.

c. $w_{n+1} - w_n = 3$: the sequence is arithmetical of common difference 3.

Personal work

Resuming Exercise 2 with the following data:
125 and rate or common difference / ratio by 3. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Session 2 — Arithmetic sequence: general term

Numerical sequences

17/09/2026 • B : 1 h – C : 2 h

Objectives of the session

· Moving from recurrence to general term.

· Calculate a remote term.

· Interpret the row.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 5 - x + 2$. $2x + 7$

2. How many terms of u_3 to u_{15} ? $15 - 3 + 1 = 13$

3. Give the coefficient of a decrease of 8 per cent. 0,92

4. Give common difference / ratio for $12, 12 + 5, 12 + 2 \times 5, \ldots$. $r = 5$

5. Calculate u_4 if $u_0 = 15$ and $u_{n+1} = u_n + 3$. $u_4 = 27$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: go from recurrence to general term and calculate a distant term.

Research activity

A store has 1,250 active loyalty cards at range 0 and earns 75 per month. all terms, determine strength at rank n, then rank 18. Explain the role of 18 in calculation.

Detailed correction · Professor edition

$u_n = 1\,250 + 75n$, therefore $u_{18} = 2{,}600$. Number 18 includes increases from grade 0. Verbate the data that triggers the choice of method.

Trace written

If (u_n) is arithmetical of common difference r, then $u_n = u_0 + nr$ or $u_n = u_1 + (n-1)r$, depending on the rank of First term.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$u_n = 1\,250 + 75n$, therefore $u_{18} = 2{,}600$. Number 18 includes increases from grade 0.

Training · essential

1. Next steps (u_n) checks $u_0 = 120$ and $u_{n+1} = u_n + 4$. Write u_n then calculate u_{12}.
2. A quantity is worth 440 and a 4% decrease per period. v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Detailed correction · Professor edition

$u_n = 120 + 4n$ and $u_{12} = 168$, For the second, $q = 0,96$ and $v_n = 440 \times q^n$; the third item is justified by the consistency of the difference or quotient,

Reference in the manual

Supplements: Calao, courses p. 18·21; exercises p. 24, n° 5·8.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 24, exercise 6

Let (l_n) the arithmetic sequence of common difference 2 and first term: $l_0 = -1$.

a. Express l_n for all natural n.

b. Calculate l_{35} and l_{64}.

Detailed correction · Professor edition

$l_n = l_0 + nr = -1 + 2n$. Thus $l_{35} = 69$ and $l_{64} = 127$.

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Session 3 — Sum of terms of an arithmetic sequence

Numerical sequences

22/09/2026 • B : 2 h – C : 1 h

Objectives of the session

· Count the terms of an amount.

· Apply the formula.

· Control the order of magnitude.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20% of 120. 24

2. Give the coefficient of a decrease of 8 per cent. 0,92

3. Give common difference / ratio for $12, 12+6, 12+2\times 6, \ldots$. $r = 6$

4. Calculate u_4 if $u_0 = 16$ and $u_{n+1} = u_n + 3$. $u_4 = 28$

5. How many terms of u_3 to u_{15} ? $15 - 3 + 1 = 13$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: count the terms of a sum and apply the formula.

Research activity

A company pays a monthly premium: 45 euros the first month, then 5 euros each calculate the total paid over the first 12 months in three ways: explicit sum, First-last grouping, formula.

Detailed correction · Professor edition

$u_{12} = 45 + 11 \times 5 = 100$. The total is worth $12 \times \dfrac{45 + 100}{2} = 870$ euros. Make the data that triggers the choice of method.

Trace written

A sum of arithmetic terms is Number of terms $\times \dfrac{\text{First} \quad + \text{last}}{2}$. From u_p to u_q, there is a $q - p + 1$ terms.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$u_{12} = 45 + 11 \times 5 = 100$. The total is worth $12 \times \dfrac{45 + 100}{2} = 870$ euros.

Training · essential

1. Next steps (u_n) checks $u_0 = 130$ and $u_{n+1} = u_n + 5$. Write u_n then calculate u_{12}.
2. A quantity is worth 460 and a 5% decrease per period. v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Detailed correction · Professor edition

$u_n = 130 + 5n$ and $u_{12} = 190$, For the second, $q = 0,95$ and $v_n = 460 \times q^n$; the third item is justified by the consistency of the difference or quotient,

Reference in the manual

Supplements: Calao, courses p. 20·21; exercises p. 24, n° 11-13, then p. 28, n° 43·44. reproduced in this session is identified below by its page and number; the reference allows also find the figures and digital resources of the manual.

Manuel Calao · p. 24, exercise 9

a. Calculate the sum of the first 500 non-zero integers $\sum_{i=1}^{500} i$.

b. Calculate the sum of the totals from 35 to 150: $\sum_{k=35}^{150} k$.

Detailed correction · Professor edition

a. $\sum_{i=1}^{500} i = \frac{500 \times 501}{2} = 125\,250$.

b. There are $150 - 35 + 1 = 116$ terms, so the sum is 116 $\times \frac{35 + 150}{2} = 10\,730$.

Personal work

Resuming Exercise 2 with the following data:
135 and the rate or common difference / ratio by 5. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.

Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Session 4 — Geometric Sequence: recognize and model

Numerical sequences

24/09/2026 • B : 1 h – C : 2 h

Objectives of the session

· Recognize multiplicative evolution.

· Switch from a rate to a coefficient.

· Compare arithmetic and geometric.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient with an increase of 25 per cent. $1 + \frac{25}{100} = 1,25$

2. Give common difference / ratio for $12, 12 + 7, 12 + 2 \times 7, \ldots$. $r = 7$

3. Calculate u_4 if $u_0 = 17$ and $u_{n+1} = u_n + 3$. $u_4 = 29$

4. How many terms of u_3 to u_{15} ? $15 - 3 + 1 = 13$

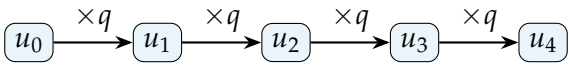
5. Give the coefficient of a decrease of 8 per cent. 0,92

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: recognize a multiplicative evolution and pass a rate at a coefficient.

Research activity

An online shop increases its number of visitors by 6% per month from 8,000 visitors. Calculate the first three values. Compare differences then quotients. Identify nature I'll do it later.



Detailed correction · Professor edition

The coefficient is 1 ,06 : 8 000, 8 480, 8 988,8. Quotients are worth 1 ,06 ; the sequence is geometric. verbalize the data that triggers the choice of method.

Trace written

A sequence is geometric when $\frac{u_{n+1}}{u_n} = q$ is constant (non-zero terms). constant leads to a geometric sequence.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

The coefficient is 1.06: 8,000, 8,480, 8 988.8. The quotients are 1.06; the sequence is geometric.

Training · essential

1. Next steps (u_n) checks $u_0 = 140$ and $u_{n+1} = u_n + 6$. Write u_n then calculate u_{12}.
2. One quantity is worth 480 and decreases by 6 per cent per period. v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Detailed correction · Professor edition

$u_n = 140 + 6n$ and $u_{12} = 212$, For the second, $q = 0,94$ and $v_n = 480 \times q^n$; the third item is justified by the consistency of the difference or quotient,

Reference in the manual

Supplements: Calao, courses p. 23·25; exercises p. 25, n° 14·18.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 25, exercise 19

Let (l_n) the geometric sequence of common ratio 2 and first term: $l_0 = -1$.

a. Express l_n for all natural n.

b. Calculate l_{12} and l_{20}.

Detailed correction · Professor edition

$l_n = l_0 q^n = -2^n$. Thus $l_{12} = -4,096$ and $l_{20} = -1\,048\,576$.

Personal work

Resuming Exercise 3 with the following data:
140 and the rate or common difference / ratio by 6. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.

Differentiation: provide fragile students with a four-box card (data, model, calculation,
To ask the fast students to produce a counter example, justification or
a second method.

Session 5 — Geometrical sequence: general term and sum

Numerical sequences

29/09/2026 • B : 2 h – C : 1 h

Objectives of the session

· Write the general term.

· Calculate a geometric sum.

· Interpret a cumulative evolution.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 8 - x + 2$. $2x + 10$
2. Calculate u_4 if $u_0 = 18$ and $u_{n+1} = u_n + 3$. $u_4 = 30$
3. How many terms of u_3 to u_{15} ? $15 - 3 + 1 = 13$
4. Give the coefficient of a decrease of 8 per cent. 0,92
5. Give common difference / ratio for $12, 12 + 8, 12 + 2 \times 8, \ldots$. $r = 8$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: write the general term and calculate a sum geometrical.

Research activity

A capital of 250 000 Ar is invested at 4% per year, compound interest. C_n after n years. C_6, then the sum $C_0 + \cdots + C_6$ and explain what the latter sum is mathematically represents and why it is not the final capital.

Detailed correction · Professor edition

$C_n = 250\,000 \times 1{,}04^n$ and $C_6 \approx 316\,330$ Ar. The sum is the sum of seven successive states, not one Real accumulation of the placement. Verbate the data that triggers the choice of method.

Trace written

If (u_n) is geometrical of common ratio q, $u_n = u_0 q^n$ or $u_n = u_1 q^{n-1}$. For $q \neq 1$, $u_0 + \cdots + u_n = u_0 \dfrac{1 - q^{n+1}}{1 - q}$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$C_n = 250\,000 \times 1{,}04^n$ and $C_6 \approx 316\,330$ Ar. The sum is the sum of seven successive states, not one actual accumulation of the investment.

Training · essential

1. Next steps (u_n) checks $u_0 = 150$ and $u_{n+1} = u_n + 2$. Write u_n then calculate u_{12}.
2. A quantity is 500 and a decrease of 2 % per period. v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Detailed correction · Professor edition

$u_n = 150 + 2n$ and $u_{12} = 174$, For the second, $q = 0,98$ and $v_n = 500 \times q^n$; the third item is justified by the consistency of the difference or quotient,

Reference in the manual

Supplements: Calao, courses p. 25·28; exercises p. 25, n° 19-25, and p. 28, n° 42·44. reproduced in this session is identified below by its page and number; the reference allows also find the figures and digital resources of the manual.

Manuel Calao · p. 25, exercise 22

What's next? (u_n) geometrical common ratio 3 and first term: $u_0 = 5$.

a. Express u_n for all natural n. Deduct from it u_7.

b. Calculate: $S = \sum_{i=0}^{7} u_i = u_0 + u_1 + \cdots + u_7$.

Detailed correction · Professor edition

$u_n = 5 \times 3^n$, therefore $u_7 = 10{,}935$. $S = 5\,\dfrac{1 - 3^8}{1 - 3} = 16\,400$.

Personal work

Resuming Exercise 2 with the following data:
145 and the rate or common difference / ratio by 7. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Session 6 — Assessment and formative evaluation

Numerical sequences

01/10/2026 • B : 1 h – C : 2 h

Objectives of the session

· Choose the right model.

· Mobilize formulae.

· Analyze errors.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 180. 9

2. How many terms of u_3 to u_{15} ? $15 - 3 + 1 = 13$

3. Give the coefficient of a decrease of 8 per cent. 0,92

4. Give common difference / ratio for $12, 12 + 9, 12 + 2 \times 9, \ldots$. $r = 9$

5. Calculate u_4 if $u_0 = 19$ and $u_{n+1} = u_n + 3$. $u_4 = 31$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: choose the right model and mobilize formulas.

Research activity

Classify eight situations according to three categories: arithmetic, geometric, neither one nor the other.
each situation, highlight the word or data that justifies the choice, then choose the appropriate formula.

Detailed correction · Professor edition

A constant absolute variation indicates the arithmetic model; a percentage or coefficient
constant indicates the geometric model; otherwise neither of the two is guaranteed.
data that triggers the choice of method.

Trace written

Before calculating, identify: type of sequence, first row, common difference / ratio, term or amount requested.
conclusion specifies unity and rank.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Detailed correction · Professor edition

A constant absolute variation indicates the arithmetic model; a percentage or coefficient
constant indicates the geometric model; otherwise neither is guaranteed.

Training · essential

1. Next steps (u_n) checks $u_0 = 160$ and $u_{n+1} = u_n + 3$. Write u_n then calculate u_{12}.
2. A quantity is 520 and a decrease of 3% per period. v_n and calculate v_5.
3. For each sequence, specify whether the model is arithmetic or geometric and justify by a difference or quotient.

Detailed correction · Professor edition

$u_n = 160 + 3n$ and $u_{12} = 196$, For the second, $q = 0,97$ and $v_n = 520 \times q^n$; the third item is justified by the consistency of the difference or quotient,

Reference in the manual

Supplements: Calao, p. 29, No 49-52. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 29, exercise 49

For each proposal, state whether it is true or false. Justify the answer.

1. A capital of 5,000 € is placed on a simple interest account at the annual interest rate of This situation can be modelled by a geometric sequence of common ratio 1,04.
2. The number of sparrows declines by 20% per year in one region. reintroduction of 200 sparrows each year. This situation can be modelled by a follow-up arithmetical common difference 200.
3. The mass of household waste falls by 10% per year. This situation can be modelled by a geometric sequence of common ratio 1.1.
4. The production of a company increases weekly by 200 items produced. This situation can be modelled by an arithmetic sequence of common difference 200.
5. The number of subscribers to a youtuber increases by 15% every month. be modelled by an arithmetic sequence of common difference 15.
6. The annual rent of an apartment increases by 320 € Every year, this situation can be be modelled by an arithmetic sequence of common difference 320.
7. On 1 1st January of a year, Enzo places 1 000 € with compound interest on an account the rate of which annual is equal to 2 % and 1 1st January of each following year, it places 200 € on this account. This situation can be modelled by a geometric sequence of common ratio 1,02.

Detailed correction · Professor edition

The answers are: false, false, false, true, false, true, false. To simple interests, the increase is constant; in questions 2 and 7, an additive operation adds to multiplicative evolution ; a decrease of 10 % corresponds to 0 ,9; an increase of 15% leads to a geometric sequence of ratio 1.15.

Personal work

Resuming Exercise 3 with the following data:
150 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.